

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

■ Operating Systems ■ CS-3009 ■ Autumn–2024, Mid Sem

■ 30 Marks ■ Time: 2hrs ■ 1 Page ■ Answer ALL Questions

Ensure that all parts of a question are answered sequentially.

1. (a) How many states a process must pass through in its life-cycle to complete? Can a process transition from waiting for an I/O operation to the terminated state? Why or why not? [2]
- (b) What advantage does indirect communication provide over direct communication? [2]
- (c) Why might threads be particularly useful in writing a server, for example, a server that provides access to items from a database? [2]
- (d) What is the effect if the time slice is chosen to be too short or too long in round-robin scheduling? Briefly explain. [2]
- (e) Differentiate between Short-term and Medium-term schedulers. Which scheduler is responsible for selecting a good mix of I/O-bound and CPU-bound processes? [2]
2. (a) Consider the set of seven processes whose arrival time and burst time are given in Table 1. Assume that the OS follows SRTF scheduling policy. Show the Gantt chart depicting the schedule of execution of the processes. Show the content of the ready queue at each scheduling point. Find the waiting time of each process. [4]

Table 1: Set of Processes

Process	AT	BT
P1	0	6
P2	0	3
P3	1	4
P4	2	2
P5	3	1
P6	4	5
P7	6	2

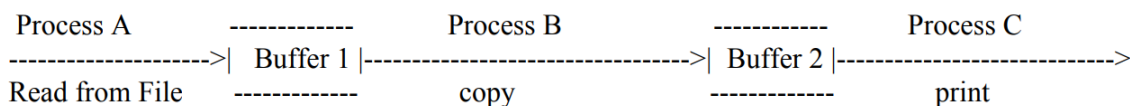
AT: Arrival Time
BT: Burst Time

Process P_i :

```
do {
    flag[i] = TRUE;
    turn = i;
    while (flag[(i+1)%2] && turn == (i+1)%2);
    <critical section>
    flag[i] = FALSE;
    <remainder section>
} while (TRUE);
```

- (b) Consider the solution to the critical section problem for two processes $P_i; 0 \leq i < 2$. The general structure of process P_i is shown above. Verify all the required criteria of a feasible solution to the synchronization problem. Corroborate your answer with a proper explanation. [6]
3. (a) Suppose that a process scheduling algorithm favours those processes that have used the least processor time in the recent past. [4]
 - Will this algorithm favour I/O-bound processes over CPU-bound processes? Justify.
 - Will this algorithm starve CPU-bound processes? Justify.

- (b) Consider a file (**F**) on the disk having a set of items. There are three processes in the system. The first process (**A**) reads items from the file and writes them to *Buffer1*. The second process (**B**) copies items from *Buffer1* to *Buffer2* and the third process (**C**) finally reads the items from *Buffer3* and prints them. The flow is shown in the following image.



Each of the buffers has the capacity of 1 item. Write and explain the synchronization code for the three processes using semaphores. [6]